Unit 6

UNITARY METHOD

6.1 UNITARY METHOD

6.1.1 Describe the concept of unitary method

We use mathematics in our daily life problems, in which the price of a number of articles is given. We are required to find the price of some other number of articles of same kind. We solve such problems by finding the price of one article.

The method in which the value of a number of articles as unit is determined by finding a value of an article is called unitary method.

6.1.2 Calculate the value of many objects of the same kind when the value of one of these objects is given

Case I:

To find the value of many objects we just multiply the value of one object with required number of objects.

Example 1.

The price of a book is **Rs 30**. Find the price of **4** such books.

Solution:

The price of one book = 30 rupees
The price of 4 such books = (30 x 4) rupees
= 120 rupees

Example 2. The price of one pencil is **Rs 4.50**. What is the price of **5** such pencils?

Solution:

The price of 1 pencil = 4.50 rupees
The price of 5 such pencils = (4.50 x 5) rupees
= 22.50 rupees

Case II:

To find the value of one object when the value of many objects is given, we divide the given value of the objects by the number of objects.

Teacher's Note Teacher should explain the concept and gives the daily life problems regarding unitary method.

Example 3.

The price of **6 kg** apples is **Rs 240**. What is the price of **1 kg** apples?

Solution:

The price of 6 kg apples = 240 rupees

The price of 1 kg apples $= \frac{240}{6}$ rupees $= \frac{\cancel{240}^{40}}{\cancel{6}_{1}}$ rupees

= 40 rupees

EXERCISE 6.1

1. The price of a pen is **Rs 30**. Find the price of **6** such pens.
2. One litre of petrol price is **Rs 105.70**. What is the price of **5 litres** of petrol?
3. The price of 1 kg rice is **Rs 110**. What is the price of **7 kg** rice?
4. The rent of a house for one month is **Rs 5000**. What is the rent for one year?
5. The price of a dozen bananas is **Rs 66**. What is the price of 1 banana?
6. If **12** books price **Rs 480**. What is the price of one book?
7. The price of **8** chocolates is Rs **36**. What is the price of one chocolate?
8. The price of ten balls is Rs **205**. What is the price of one ball?
9. The price of three mobile phones is Rs **38400**. What is the price of one?
10. The price of **16** pairs of socks is Rs **1004.80**. What is the price of one pair of socks?

6.1.3 Solving to word problems when the same type of objects are given (unitary method).

If value of many things is given. First we find the value of one thing (by division) and then find the value of the required number of things. (by multiplication).

Example 1.

Rafay purchased **5** copies for **Rs 100**. How much will he pay for **12** such copies?

Solution:

The price of 5 copies = Rs 100

The price of 1 copy = Rs $\frac{100}{5}$

The price of 12 copies = Rs $\left(\frac{\overset{20}{\cancel{100}}}{\underset{1}{\cancel{5}}} \times 12\right)$

= Rs (20 x 12)

= Rs 240

Hence Rafay will pay Rs 240 for purchasing 12 such copies.

Example 2.

Bismah reads **50** pages of a book in **3** hours. In how many hours can she read the book of **250** pages?

Solution:

For reading 50 pages, she takes = 3 hours

For reading of 1 page, she will take = $\frac{3}{50}$ hours

For 250 pages, she takes = $\left(\frac{3}{\underset{1}{\cancel{50}}} \times \overset{5}{\cancel{250}}\right)$ hours

= (3 x 5) hours **= 15 hours**

So, Bismah can read the book in **15 hours**.

EXERCISE 6.2

1. The price of **6** balls is **240** rupees. Find the price of **10** such balls.

2. The cost of **10** books is **Rs 240**. What will be the cost of such **15** books?

3. The price of **2** dozen pencils is **Rs 60**. What is the price of $3\frac{1}{2}$ dozen pencils?

4. **6** farmers plough a field in **10** hours. How many hours will it take for **8** farmers to plough the same field?

5. Rent of a house for **3** months is **Rs 18000**. What is the rent for **8** months?

6. A car travels **45 km** in $3\frac{1}{2}$ **litres** of petrol. How far would it travel with **364 litres** of petrol?

7. **6** metres of cloth is required for **2** shirts. How many same shirts can be made form **42** metre?

8. **12** coaches of passengers carry **624** persons. How many passengers will be carried by **18** such coaches?

9. The weight of **16** bags of rice is **775.60** kg. What will be the weight of **24** such bags?

10. The bus fare of **10** persons from Karachi to Larkana is Rs **8300**. What is the fare for 36 persons?

6.2 DIRECT AND INVERSE PROPORTION

6.2.1 Define ratio of two numbers

Ratio:

A ratio is a relationship between two numbers. The numbers represent quantities of same kind. Ratio of two numbers 3 and 2 is expressed as 3 : 2 or $\frac{3}{2}$ We read 3 : 2 as (3 is to 2)

Note: Ratio compares two quantities of same kind.

For example: The ratio of ages of Danish and Rafay is 3 : 1 then it shows that

- Danish is older than Rafay
- Danish is thrice as old as Rafay

The ratio of two quantities **a** and **b** is written as **a**: **b**. We read this as "**a** is to **b**" **a**: **b** is also written as $\frac{\mathbf{a}}{\mathbf{b}}$, where $b \neq 0$.

Proportion:

Equality of two ratios is called proportion. The symbol for proportions is "=" or "::"

If **a**:**b** and **c**:**d** are two ratios then the proportion between these two ratios is written as:

a:b = c:d or a:b :: c:d.

It is read as "Ratio **a** is to **b** is same as **c** is to **d**". Here **a**, **b**, **c** and **d** are respectively called as first term, second term, third term and fourth term of the proportion.

Example. 2 : 3 = 4 : 6 is a proportion.

We can write is as 2 : 3 :: 4 : 6

6.2.2 Define and identify direct and inverse proportion

Types of proportion:

There are two types of proportion.

(i) Direct proportion
(ii) Inverse proportion

Teacher's Note Teacher should explain the concept of ratio also describe the types of proportion with daily life examples.

(i) Direct Proportion: If two quantities are related in such a way that if one quantity increases in a given ratio, the other also increases in the same ratio or if one decreases in a given ratio, the other also decreases in the same ratio. Then the given quantities are said to be in direct proportion.

Examples:

(1) More money and more shopping; less money and less shopping.
(2) Faster the speed of machine and more production is produced.

(ii) Inverse Proportion: If two quantities are related in such a way that if one quantity increases in a given ratio, the other decreases in the same ratio. Then the given quantities are said to be inverse proportion.

Examples:

(1) Faster the speed, lesser the time taken
(2) More workers, less number of days for completing a work.

Activity **Identify and write direct or inverse proportion.**

(i) Larger crowd, more noise. (Direct proportion)
(ii) More books, more money needed. (.............................)
(iii) Less labourers, more time to build a house. (..................)
(iv) Less money, less toffees purchased. (.............................)

6.2.2 Solve real life problems involving direct and inverse proportion (by unitary method)

Example 1. A man buys **3kg** of apples for **Rs 150**. How much will he pay for **7kg** of apples?

Solution:

Kg		Rs		Kg		Rs
3	:	150	::	7	:	$\boxed{}$

For 3kg of apples he spends Rs 150

For 1kg of apples he spends Rs $\frac{150}{3}$

For 7kg of apples he spends Rs $\frac{\cancel{150}^{50}}{\cancel{3}_{1}} \times 7$ **= Rs 350**

Example 2.

8 workers can do a work in **6** hours. How much time will **12** workers take to do the same work?

Solution:

Workers		Hours		Workers		Hours
8	:	6	::	12	:	$\boxed{}$

8 workers can do the work in = 6 hours

1 worker can do the work in = 6 x 8 hours (Less workers more time)

12 workers can do the work in $\dfrac{\cancel{6}^{1} \times \cancel{8}^{4}}{\cancel{12}_{\cancel{2}_{1}}}$ hours = $\mathbf{\dfrac{4}{1}}$

(More workers less time) **= 4 hours**

So, **12 workers** will do the same work in **4 hours**.

EXERCISE 6.3

1. The ratio of pocket money of Bismah and Omaima is 3 : 5. Write true or False.

(i) Omaima has less pocket money than Bismah. ( )

(ii) Bismah has less pocket money than Umaima. ( )

2. Express the following as ratio.

(i) 3 boys to 6 girls (ii) 25 minutes to 80 minutes

(iii) 5 days to 3 weeks (iv) 250 kg to $1\frac{1}{2}$ kg

3. Identify and write direct or inverse proportion.

(i) More laboures, more work. ( )

(ii) More laboures, less days needed for work. ( )

(iii) Less speed of a car, more time taken. ( )

(iv) More farmers, more work done in the field. ( )

4. If 2 packs of juice cost **Rs 24**. what is the cost of **4** packs of juice?
5. Aslam can walk **10.5 km** in **2** hours. Find the distance he can travel in **5** hours.
6. If **9** workers can complete a work in **6** days. How many workers are required to complete the same work in **3** days?
7. A motorcycle covers **100 km** in **2** $\frac{1}{2}$ litres of petorl. How many litres of petrol will it use to cover **300 km**?
8. **180** soldiers have food stock for **6** days. How many soldiers can eat the same food for **9** days?
9. **44** farmers could reap a field in **15** days. How many farmers can reap the same field in **10** days?

REVIEW EXERCISE 6

1. The cost of a sharpener is Rs. 4.50. Find the cost of a dozen sharpeners.
2. From Karachi to Nawabshah the fare of 10 persons in a bus is 6500 rupees. What is the fare of each person?
3. The cost of 2 dozen eggs is 220 rupees. What is the cost of 3 dozen eggs?
4. If 10 persons can do a work in 6 days. In how many days can 15 persons do the same work?
5. The price of 6 packets of chalk is Rs. 90. What is the price of 8 packets of chalk?
6. A printer prints 7620 papers in 1 hour. Find how many papers will be printed by the same printer in 40 minutes?
7. A train travels 800 km in 6 hours. How much distance will be traveled in 15 hours with same speed?